

Kelvin Ching Kiu Chan

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EDUCATION

San Francisco State University – San Francisco, CA

August 2018 – May 2022

Bachelor of Science in Computer Science; Minor in Mathematics

GPA: 3.75; *magna cum laude*

Honors: Dean's List (6 semesters)

EXPERIENCE

Code Tenderloin – San Francisco, CA

August 2022 – February 2023

Freelance Web Designer

- Implemented 20+ stakeholder-requested enhancements across four iterative design revisions, addressing UI, navigation, responsiveness, and content presentation issues.
- Troubleshoot and resolve browser compatibility, responsive design, and UI issues (including image cropping, gallery ordering, navigation, and video embedding), improving overall user experience.
- Updated website sections (Team, Bookmark Exhibit, navigation, color scheme) based on client feedback, resulting in a more professional and cohesive final product.

Web Developer Intern

June 2022 – July 2022

- Developed four narrative websites using HTML5, CSS3, and JavaScript, improving usability through animations, responsive layouts, and enhanced visual presentation.
- Crafted cross-platform website wireframes on Figma, prioritizing content organization, structural planning, and UX design.
- Diagnosed and resolved cross-browser compatibility and UI issues across 4 web applications.
- Assisted and trained peers on Browser Developer Tools and mobile responsiveness, improving collaborative debugging efficiency.

PROJECTS

Job Application Tracker App – (Full Stack Web Application)

Full Stack Developer

October 2023 – January 2024

- Developed a Single Page Application using React.js, Django REST Framework, MongoDB, and Axios for job application tracking and task management.
- Implemented secure authentication with Bcrypt hashing, CSRF protection, and input validation.
- Worked in Agile development with 2-week sprints, daily stand-ups, feature branching, pull requests, and code reviews via Git.
- Used Postman for independent backend API testing to support parallel development.

Target Store Listing iOS App - (iOS Application)

Software Developer

November 2021 – December 2021

- Developed iOS application using Swift and UIKit with API integration for inventory tracking and optimized data retrieval using asynchronous requests and caching.

4List Community Marketplace – (Human-Computer Interaction Project)

Lead Frontend Developer

September 2021 – December 2021

- Collaborated in an Agile software development team to design and develop a Craigslist-inspired listing platform focused on improving usability for buying, selling, jobs, housing, and community events.
- Developed responsive front-end components using HTML and Bootstrap CSS.
- Designed usability surveys and feedback analysis, identifying opportunities for improved filtering, layout alignment, and user workflows.

Pottr.app – (Full Stack Web Application)

Lead Frontend Developer

August 2021 – December 2021

- Collaborated to develop a video monetization platform for gaming clip competitions.
- Refactored legacy React components by migrating nested Promise chains to async/await and implementing a centralized Axios API service layer to improve code organization.
- Developed custom hooks to eliminate prop drilling and modularize filtering, searching, and data loading.
- Optimized performance and navigation using React Suspense, lazy loading, code-splitting.
- Built file upload functionality supporting multiple video clips with proper FormData handling and defensive error management for API responses.

Blog Database System – (Database Systems Project)

Database Designer & Developer

March 2021 – May 2021

- Designed and implemented a normalized MySQL relational database schema using ER diagrams with proper entity relationships and constraints.
- Ensured data integrity and reliability through primary/foreign keys, NOT NULL constraints, referential integrity, and ACID-compliant design, reducing data redundancy via normalization techniques.
- Conducted comprehensive testing with complex SQL queries (joins, aggregates, subqueries, GROUP BY, HAVING) to validate data accuracy, storage, retrieval, updates, and deletion cascades.

Tank Game 2D – (Object-Oriented Programming Project)

Software Developer

June 2020 – July 2020

- Developed a local multiplayer split-screen 2D game in Java using Swing and MVC architecture.
- Implemented strategic gameplay components with OOP abstraction and polymorphism, ensuring correct mobility detection for streamlined updates.

Linux File System Emulator – (Operating Systems Project)

Lead Developer

March 2020 – May 2020

- Designed and implemented a hierarchical Linux-style file system in C with tree structures and simulated LBA.
- Utilized GDB to diagnose and resolve segmentation faults caused by invalid pointer access during directory traversal; implemented boundary checks, include guards in header files, and basic serialization logic to prevent data corruption and improve system stability.
- Managed core file operations (create, read, write, delete, move) along with import/export features (getReal and setReal) for integration with the host Ubuntu filesystem.
- Created a Makefile to simplify compilation and execution of the program

Linked Bag Data Structure Implementation – (C++ Course Project)

Software Developer

July 2019

- Implemented a templated LinkedBag container in C++ using a linked list, supporting operations including addEnd, removeSecondNode, removeRandom, and frequency counting.
- Developed both iterative and recursive solutions for size and frequency functions to compare performance and design approaches.
- Refactored raw pointer implementation to use modern C++ smart pointers (std::unique_ptr) to improve memory safety and prevent memory leaks.
- Analyzed and addressed common memory management issues such as dangling pointers and double deletion.

SKILLS

Programming: Python, C/C++, Java, JavaScript, SQL, HTML/CSS

Frameworks & Databases: React.js, Django, Django REST Framework, Bootstrap CSS, MongoDB, MySQL, REST APIs

Software Engineering: OOP, MVC, SDLC, Agile Development, Debugging, Code Reviews, Cross-Browser Testing, Responsive Design

Tools & Security: Git, Postman, Figma, Linux, GDB, Authentication, Bcrypt, CSRF Protection, API Testing

Languages: English (Native), Cantonese (Native), Mandarin (Proficient)